

the other as reference image. During the registration process, the target image is transformed in a way that it aligns with the reference image. The transformation can be done in multiple ways, for each of them are best suited to certain types of images. The registration process can try to align target and reference images based on quite a number of registration methods. Some are:

- 1 **Intensity based:** When entire portions of images can be matched.
- 2 **Feature based:** When particular parts of images such as lines, contours etc can be matched.
- 3 **Transformations based:** Rotated or scaled versions of images.
- 4 **Multi-modality based:** When images produced by different types of sensors are joined together.

NOTE

On these pages, by the word 'image', we are not referring to just the ones which we take from a camera. An image produced by a radar would not contain all the colors human eye can see but in terms of computing would still qualify as an image. Computers treat images as a map of pixels where each pixel contains some data. In case of a normal image (the ones we can take from a mobile camera), pixels contain color-data and in case of radar, pixels can be used to represent the distance of the point from the radar. Since such data can be expressed as a visible map, data received from a radar system can be used for image registration.

The above list of registration methods is not exhaustive and there are many which are given a name by combining more than one of them. e.g. spatial registration method involves intensity, feature as well as transformation based methods at the same time.

All these methods of image registrations are imperfect. Though one of the prime concerns of image registration is to reduce errors. An error in matching the patterns in the received images can result in the target image overlaid on the reference image with a planar difference between two points which represent the same part of image (it would be appear as if a displaced layer of the same image has been put over the actual one).

Right from merging more than one images of the same object to showing more than one object on screen, this technology plays a role in recognition based AR. If the images received from more than one sensors are not merged properly, the application fails to serve its purpose.